



SERUM ROUTING TOOL DEVELOPMENT AT THERMO FISHER SCIENTIFIC

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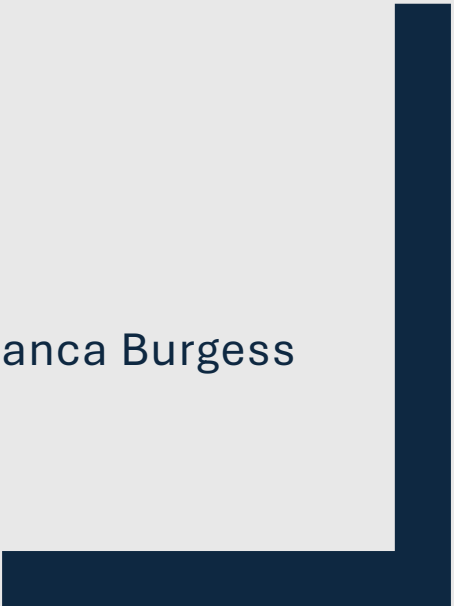


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Executive Summary

During my time at Thermo Fisher Scientific, I received many assignments but my main focus throughout my yearlong internship was updating the routings for the Serum products. Routings set the time standard for how long it takes to make a product depending on the accounting cost quantity (ACQ) which is the batch size of a job. These serum routings have not been updated in 20 years and when they were made, they were inaccurately produced using improper methods. It is important that these routings are updated as these routings affect the cost to make the product.

After conducting time studies and breaking down the data, major differences were found in over 100 serum SKUs. For example, for one SKU we found an 11 hour increase from the current to the future routing which means there was a \$3.57/liter increase from the current costing to the future costing. From the data that was collected, we were able to produce a serum routing tool for both bulk and finished good serum SKUs. The routing tool will automatically generate the BOM of the product and the updated routing based on the ACQ, my data and the BOM.

From the data collection and development of the serum routing tools, improvements have been made for over 100 serum SKUs. This not only means more accurate time standards on the processes, but it also means more accurate costing of the products. Additionally, the serum routing tools will benefit SKU coordinators as they will be able to automatically generate the updated routings and BOM for any serum SKU. Lastly, some conclusions that were made from doing the time studies is that the process is very manual, and it is difficult to standardize a process. Thus, I would recommend using this data to support continuous improvement products for production of serum.

Introduction

Thermo Fisher Scientific is a global medical supply company formed in 2006 from the union of Thermo Electron Corporation and Fisher Scientific International Inc. The Grand Island location also known as GIBCO focuses on making cell culture media, which is essentially “food for cells” which is used for cell growth in research settings. There are two forms of cell culture media that is made at Thermo Fisher, Liquid and Powder. Each of these forms of cell culture media are divided by their material types, Animal Origin (AO) and Animal Origin Free (AOF). During my

internship, I focused on Serum which is an AO liquid product. Serum was one of first products produced at GIBCO as the facility used to have a farm where they used the animals for serum production. It continues to be one of the most lucrative products due to its high product value.

Current routings for the sera product portfolio are outdated and inaccurate causing challenges for cost accounting and production planning. These routing have not been updated in 20 years and when they were created, it was done using improper methods. Thus, I was chosen to work closely with two other industrial engineers to collect data and update the serum routings for more accurate time standards and costing.

Statement of Engineering Problem

The routings for serum products have been inaccurate and outdated for the past 20 years. There were several issues of them not aligning with the current process, missing departments in the routing sheets, and not providing accurate time standards. As a result of this, we are not getting accurate costing to make the products.

Thus, several objectives must be completed to have more accurate routings and to create the routing tool. The first one is to create a process map and learn what are the necessary steps to complete the process.



Figure 1: Serum Process Map

Next, we had to film the time studies in each of the departments and learn more about how they all flowed into each other.



Figure 2: Photo Taken During Time Study of Prefiltration Process

After the time studies were collected, the data was taken from the videos and converted into routings. From there, we are able to make the serum routing tools in excel and incorporate the data into the tools.

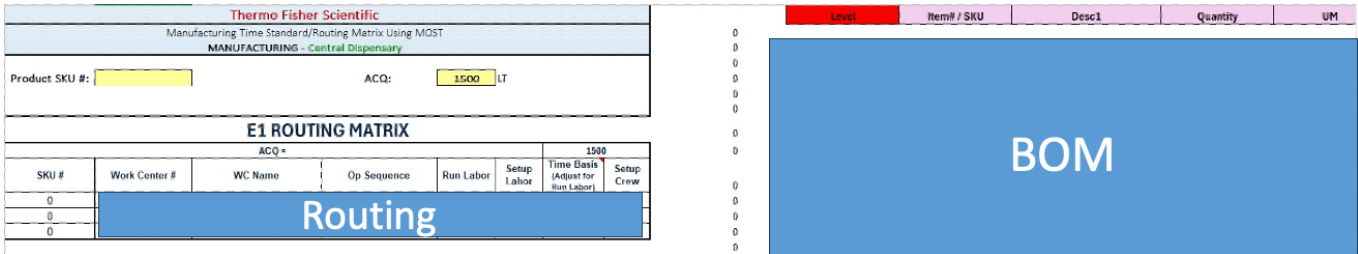


Figure 3: Serum Routing Tool

Engineering Standards, Codes and Constraints

The following are all the relevant standards, regulations, and codes that had to be followed during my time at Thermo Fisher Scientific:

- **Clean Room gowning requirements (ISO 14698-1:2003):** It is a requirement to undergo the proper gowning procedure before entering the filtration suite. If you are not certified, someone who is certified must accompany you into the room and walk you

through the gowning procedure. This influences my work as I am required to go into the filtration suites to film my time studies. Thus, I had to be accompanied every time I wanted to enter the clean room.

- **Steel toe shoe requirement (ISO 20345:2021):** It is required that all employees must wear steel toe shoes on the floor to prevent injury. This directly impacts my work as I am frequently on the floor filming time studies so I must always have proper shoes on before I go to film time studies.
- **AO/AOF Segregation:** The facility is divided into two parts, AO and AOF, with different color scrubs, teal and blue respectively. It is important to maintain the segregation by not wearing the teal scrubs in the AOF area and vice versa. This is relevant with my work as I had to make.

The following are constraints that influenced the nature of my solution approach:

- **Certain processes are run at specific times of the day:** When it comes to the production of serum, the main production begins on Tuesdays and Thursdays at 3 am. Thus, there were time that I would have to go in early in the morning to film time studies. Additionally, I would have to block out time in my day to film time studies that would only occur at specific times in the day.
- **It is different jobs every day:** The facility is a 24/7 facility and different products are made very often. Thus, if you really want to see a specific job, you had to make it your priority and make time for it. It was especially challenging when production is running behind, so you are often left waiting for it to begin.

Development of the Engineering Approach

When working on the serum routing project and developing the tool, I had to understand the scope of the problem and gain more background about the serum process. This was done by talking to other professionals who work closely in serum and related departments, reading over SOPs and WINs, and looking at previously filmed videos. Once I had a good understanding of the process, I created a high-level process map that showcases the key steps in the serum production process and made a list of all the necessary time studies that needed to be performed.

Then, we began performing time studies in all the key departments which was done by filming the process to review at our desks. Issues occurred during this as we had to coordinate with team leads when it was a good time to go down to the floor and film the process. It became more difficult as some processes were done outside of normal work hours which led me to coming into work at 3 am several times to perform time studies. Once the time studies were filmed, we reviewed the videos and extracted key data from the time studies that would be implemented into the routing tools.

During the development of the serum routing tool, some challenges arose. One challenge was figuring out whether the data was considered run labor or setup labor. The difference between the two is that run labor is variable (based on the ACQ) while setup labor is something that is done every single time and does not depend on the ACQ. Additionally, we came into issues of where to cost certain processes. This is because each department have different costs so placing the routing in a more expensive department will increase the cost for the company to produce the product. To overcome this challenge, my team and I worked closely together and met frequently to problem solve these issues.

Another problem that occurred was changes in the process. There were two changes that occurred, which were the HI process and the peel and relabel process. The heat inactivation process (HI) was affected because new water baths were installed. For the peel and relabel process, which is mainly used for irradiated serum, had a change in its process where operators are no longer needed to peel and relabel. They can simply place a sticker on that indicates that it's an irradiated product. By which led to me performing another set of time studies on these processes and updating the routing tool.

Through all these challenges, we were successfully able to create a routing tool that automatically generates accurate serum routings.

The engineering methods that were used during this project include:

- **Process Mapping:** I used process mapping during my internship to help me learn about the flow between departments during the production of serum (IE409-Six Sigma).
- **Time Studies:** This was the main way I was able to collect data for the serum routing tools (IE323- Human Factors).

- **Time Standard Calculations:** I had to calculate the basic and standard time of the processes (IE323-Human Factors).

Activities and Analysis

To develop the serum routing tool, we first had to perform time studies. This was done by filming the tasks that are being completed by the operators. To obtain data from this, we reviewed the video to determine how long it took to complete each task in a process. The raw data collection is shown below:

Pre-Filter train setup			Prefiltration			4.5L Dump			Packaging 500ml bottles			Prefiltration Disassembly		
Crew Size 1.00 person			Crew Size 1.00 Person						Crew Size 1 person			Crew Size 1 Person		
Stage I Filtration			Prefiltration Setup						Packaging one box			Stage		
Setup of filter 1	min	Data	Pooling Tank Rise	min	Data	Dump one bottle of Sera	min	Data	Packaging one box	min	Data	Part 1	min	Data
Hoisting of filter 1	min		Tank Hook Up	min		Dump one bottle of Sera	min		Packaging one box	min		Part 2	min	
Setup of filter 2	min		Pump Set Up	min		Dump one bottle of Sera	min		Packaging one box	min				
Hoisting of filter 2	min		Batch record review	min		Dump one bottle of Sera	min		Packaging one box	min				
Total Time on Stage I	min					Dump one bottle of Sera	min		Packaging one box	min				
Stage II Filtration			Move from Thaw Room						Palletizing one box					
Filter 1		Data	Move from Thaw Room	min	Data	Dump one bottle of Sera	min	Data	Palletizing one box	min	Data			
Filter 1 set up/unpackaged	min		Move from Thaw Room	min		Dump one bottle of Sera	min		Palletizing one box	min				
Place the housing 1	min		Move from Thaw Room	min		Dump one bottle of Sera	min		Palletizing one box	min				
Flush pt1	min		Move from Thaw Room	min		Dump one bottle of Sera	min		Palletizing one box	min				
Flush pt2	min		Average Move from Thaw Room	min		Dump one bottle of Sera	min		Palletizing one box	min				
Total Flush	min					Dump one bottle of Sera	min		Palletizing one box	min				
Total Filter 1	min		Total Move from Thaw Room	min	Data	Dump one bottle of Sera	min		Palletizing one box	min				
Filter 2		Data	Prefiltration Process			Dump one bottle of Sera	min	Data	Palletizing one box	min	Data			
Filter 2 set up/unpackaged	min		Change the Cloth	min	Data	Dump one bottle of Sera	min		Palletizing one box	min				
Place the housing 2	min		Total Change of Cloth	min		Dump one bottle of Sera	min		Palletizing one box	min				
Flush pt1	min		Room Cleaning	min		Dump one bottle of Sera	min		Palletizing one box	min				
Flush pt2	min		Change Dial for Free	min		Dump one bottle of Sera	min		Palletizing one box	min				
Total Flush	min		Total Changing Dial	min		Dump one bottle of Sera	min		Palletizing one box	min				
Total Filter 2	min		Stage II Pre-Filtration	min		Dump one bottle of Sera	min		Palletizing one box	min				
Filter 3		Data	Fill up Nalgene	min	Data	Dump one bottle of Sera	min	Data	Palletizing one box	min	Data			
Filter 3 set up/unpackaged	min		Pxi Calibration	min		Dump one bottle of Sera	min		Palletizing one box	min				
Place the housing 3	min					Dump one bottle of Sera	min		Palletizing one box	min				
Flush	min					Dump one bottle of Sera	min		Palletizing one box	min				
Total Filter 3	min					Dump one bottle of Sera	min		Palletizing one box	min				
Final Filtration		Data			Data	Dump one bottle of Sera	min	Data	Palletizing one box	min	Data			
Final Filter set up/unpackaged	min					Dump one bottle of Sera	min		Palletizing one box	min				
						Dump one bottle of Sera	min		Palletizing one box	min				
						Dump one bottle of Sera	min		Palletizing one box	min				
						Average Dump of one bottle of	min		Total Average time on packaging	min				

Figure 4: Data Collected from Time Studies

Then these times were transferred to the serum routing tool where we categorized everything by operation center along with determining if it is considered set up or run labor. If it was run labor, we also had to determine the time basis of the process. This was determined based off the run labor time along with the ACQ. Additionally, we had to determine the basic and standard times of the tasks, and we had to make sure the final times were in hours. To determine basic time, we multiplied the frequency of it occurring by the time it took by the crew size. Then, to find the standard time we took the basic time and multiplied it by one plus the contingency percentage (10%) which was all divided by the performance level percentage (75%). The following is an example of what the completed data looks like in routing tool:

500ML-HI							Contingency	10.00%
							Performance Level	75%
Setup Labor	Frequency	Time taken(min)	Crew Size	Basic Time	Standard Time			
Warm Up Water Bath(10 baths)	Data							
				SETUP LABOR	Data	MINS		
						HOURS		
Run Labor	Frequency	Time taken(min)	Crew Size	Basic Time	Standard Time			
Load bottles into 1 Water Bath	Data							
Setup for unloading								
Temp Check(1 bottle)								
Paperwork								
Time in Water Bath								
Final Temp Check								
Unloading water bottles								
Unloading water bath								
Move Racks to doors								
Finishing Run Time/bottle								
Move the Pallet Jack								
Move the Pallet Jack to alley								
RUN LABOR PER BATCH					RUN PER LITER	TIME BASIS		RUN LABOR
						HOURS		
Data								

Figure 5: Data Broken Down for Routing

Finally, we had to connect the data the background and bring it to the “front-end” of the routing tools. This was done through creating excel formulas where it would pull specific information based on the ACQ, BOM, size of bottle, and if it is a specialty serum. Below, shows an example of what the tools look like:

Thermo Fisher Scientific						Level	Item# / SKU	Desc1	Quantity	UM
Manufacturing Time Standard/Routing Matrix Using MOST						BOM				
MANUFACTURING - Central Dispensary										
Product SKU #:		ACQ: 1500 LT								
E1 ROUTING MATRIX										
ACQ *										
SKU #	Work Center #	WC Name	Op Sequence	Run Labor	Setup Labor	Time Basis (Adjust for Run Labor) Setup Crew				
0		Routing								
0										
0										

Figure 6: Serum Routing Tool

From the work I have done, for one particular SKU we found an 11 hour increase from the current to the future routing which means there was a \$3.57/liter increase from the current costing to the future costing. This tool is will be helpful for cost accounting and SKU coordinators to automatically generate routings that are accurate and up to date.

Actions Taken/Recommended

From the serum routing tool and the data that I have collected, improvements can be made to the production of serum.

A way that the data that I have collected has been useful was for proving an improvement between the older water baths and the new water baths in serum. The water baths are used for heat inactivating (HI) serum products. Our original data consisted of times from the old heat inactivation process using the old water baths. New water baths were installed in January 2025, thus new data has been collected as each bottle size gets validated. We were able to prove a 30-minute time savings as the operators are not required to stand by the water baths while the serum is getting heat inactivated. This not only means time savings as the operators are able leave and do other tasks, but this would also translate into cost savings as well.

Another way they can use our data to make improvements to the serum production process is by implementing new tools that will make it more ergonomic while also saving time and money. One tool that would be useful is a decapper for the raw material. The raw sera come in bottles ranging from 3 to 6 L in volume. There is a process during prefiltration which is called “dumping” where operators must manually unscrew the lid, dump out the liquid and place the container back on the shelf. If they are running a job that is 2000L, which is common, they would be doing this task 333 to 667 times.

Therefore, a decapper would be a useful tool that would provide ergonomic support but may help with reducing time spent on a process and standardize it which would improve cost savings. If they bought a decapper, they could run another time study and compare the number of minutes it takes to do a 2000L job against the data I have collected. Also, RULA tests could be performed comparing the result with and without the decapper. There is a possibility that the decapper isn’t going to provide any improvement in time, cost and/or ergonomics. Thus, I would suggest increasing the crew size for the tasks to take the ergonomic burden off the operator as it is typically only done by one person. It also may help time spent on the process go by faster which would save time and money.

Lastly, they can use our data to make improvements to the serum production is by implementing tools to help standardize the process. This was noticed with one operator that I time studied

where he would torque the bottles differently than the rest of the operators. This operator's way was not only an incorrect method but also was significantly faster than the other operators. It would take this operator about 3 seconds to torque one bottle while it took the rest about 7 seconds. The operator also explained that the reason why he would do their method is that normal way caused a lot of wrist strain. Therefore, the process is not standardized if one person is doing it differently than the rest and it is causing workers wrist pain.

From this, I would suggest investing in a torquing tool for the operators to reduce wrist pain and standardize the process. There are manual and automatic ones to choose from. This tool would not only make ergonomic improvements but with the standardization of the process, it would improve time and cost savings as everyone would be doing it the same way. Additionally, implementing this tool may be faster than the original method. Similarly to the decapper, they can perform time studies and RULA tests and compare the original method against the tool. They would be able to compare minutes the process takes and see if there are any dollar savings with the new tool. There is the possibility that the torquing tool won't fit the cap size which I suggest manufacturing adaptations to make it fit. For example, if the handle on the tool works, you could get a wrench part manufactured to fit the cap size.

Appendix: Self-Assessment

- a. The professional traits that are most valued by employers that I selected as areas of focus include:
 - a. Detail Orientated: I would like show my employers that I pay attention to every detail while completing my work
 - b. Teamwork: I would like to demonstrate that I am able to work well with others
 - c. Initiative: I would like to show that I am capable to doing things on my own and allow my creativity to come through
 - d. Good Communication: I want my employers to feel that they know what I am doing at any point my projects and feel up to date on what is happening
 - e. Reliability: I want to be able to hit my deadlines
- b. The professional traits that I would say that I clearly demonstrated during my internship would be
 - a. Detail Orientated: Throughout my internship, I have had to pay attention to every detail when it came to developing the new routings for serum SKUs. This was exemplified when looking at the data I collected, I had to make sure that the data I was including in the routing was aligned with the process. Additionally, I had to make sure I was not missing any steps in the process while developing the routings. If I was, I had to make sure I collected the data for it.
 - b. Teamwork: For all of my projects during my time at Thermo Fisher Scientific, I had to work with at least one other people. This is shown when working on the routings for serum routings. I had weekly meetings with the IE team to give updates on my work and get their consultation on my work. Additionally, I had to work with people on the floor to understand the process more deeply and consult them on my data. This helped me develop a strong relationship with everyone I worked with.
 - c. Initiative: I took it upon myself to learn more about an issue and not wait for my manager to tell me what to do. This was exemplified when I created a map showing how all the SKUs were connected. Making the map was not an

assignment that I had to do but more of an initiative I took to benefit me but also people who wanted a deeper understanding of how the SKUs are connected.

- d. Good Communication: I openly communicated what I was working on not only with my manager but the whole department. This is shown as I do not wait until my 1 on 1 or team meetings to address issues and give updates. I happily go up to my manager and coworkers and discuss it.
- e. Reliability: I am able to be a reliable resource for things such as data and I always gave my work in on time. This is shown as my coworker needed data to prove that the new water baths have time and cost savings. I was able to gather and prepare data in a readily fashion for her.

The following I am willing for the instructor to share the video.

- c. ABET accreditors
- d. Current and past ISE Dept Chair
- e. ISE Undergraduate Director
- f. ISE faculty
- g. ISE students